D3.3 Homeostasis: data-based questions

NAME	DATE	TOTAL
		_____ / 19

Read the data carefully before you answer. The number of marks shows how many separate points to make. Show your working in calculations.

1 A glucose tolerance test

7 MARKS

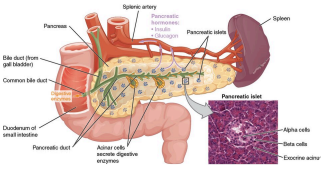
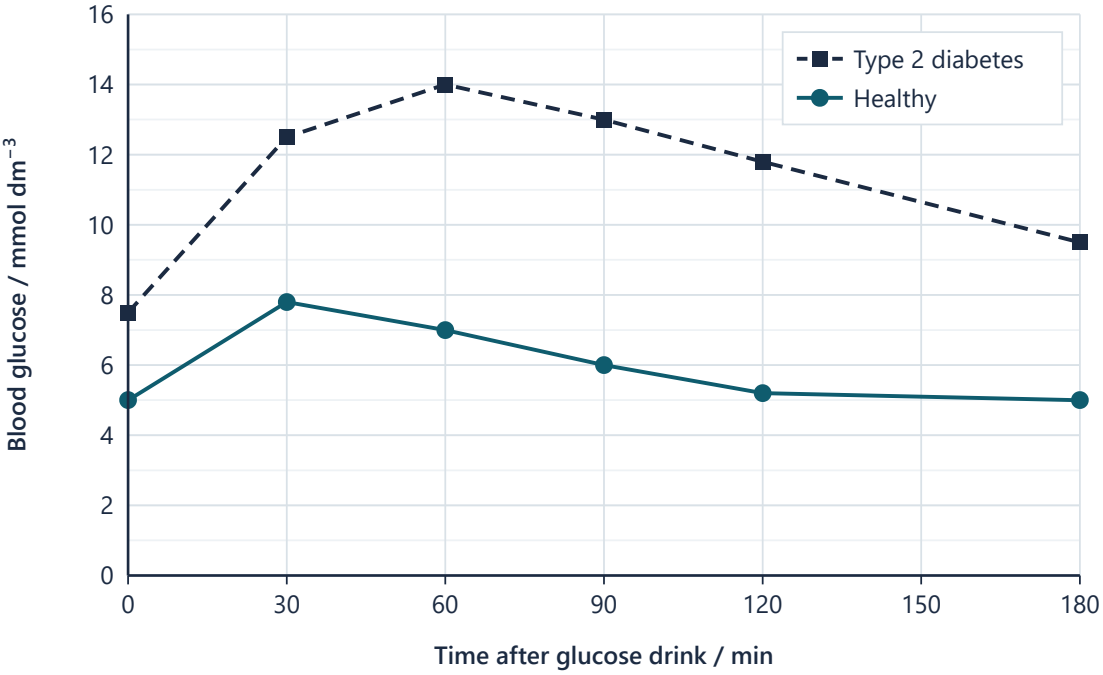


Image: OpenStax College, CC BY 3.0, Wikimedia Commons.

In an oral glucose tolerance test, a person who has not eaten overnight drinks a solution containing 75 g of glucose, and blood glucose is measured for three hours. The graph shows mean results for 20 healthy adults and 20 adults with type 2 diabetes.



a STATE the blood glucose of healthy adults 60 minutes after the drink. [1]

b COMPARE the results of the two groups. [2]

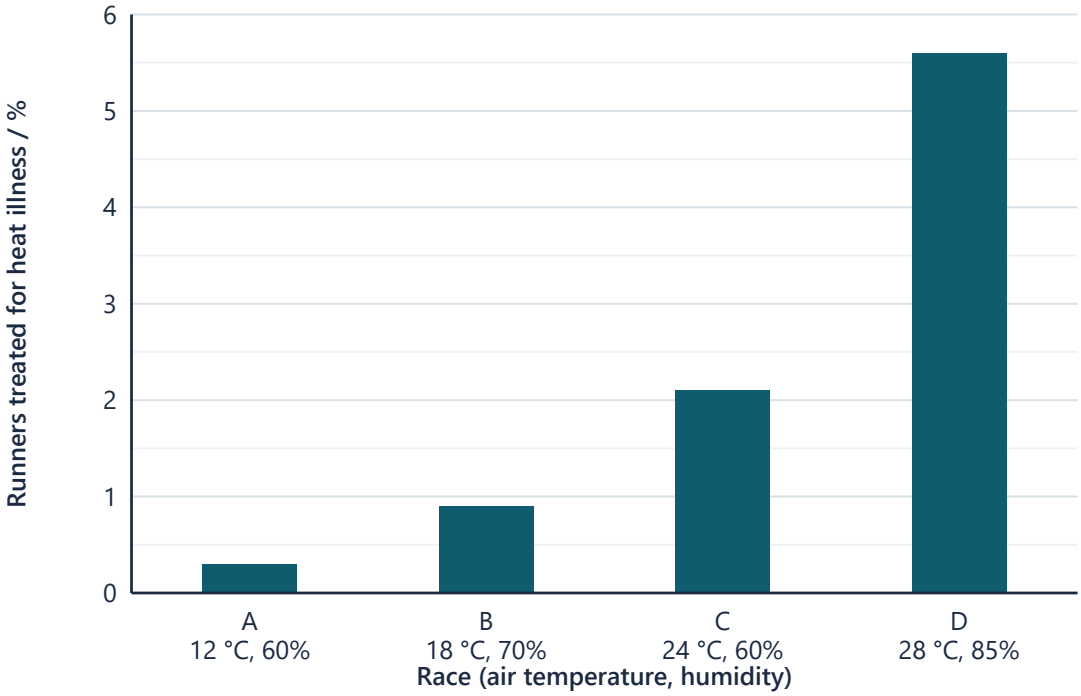
c A value of 11.1 mmol dm^{-3} or more at 120 minutes indicates diabetes. **Deduce** whether the test identifies the diabetic group. [1]

d EXPLAIN how blood glucose returns to normal in the healthy adults. [3]

2 Heat illness in marathon runners

6 MARKS

Organisers of four city marathons recorded the percentage of runners treated for heat illness. Weather at the start of each race is shown.



a DESCRIBE the relationship between air temperature and heat illness. [1]

b One organiser claimed that humidity, not temperature, causes most heat illness. EVALUATE this claim using the data. [2]

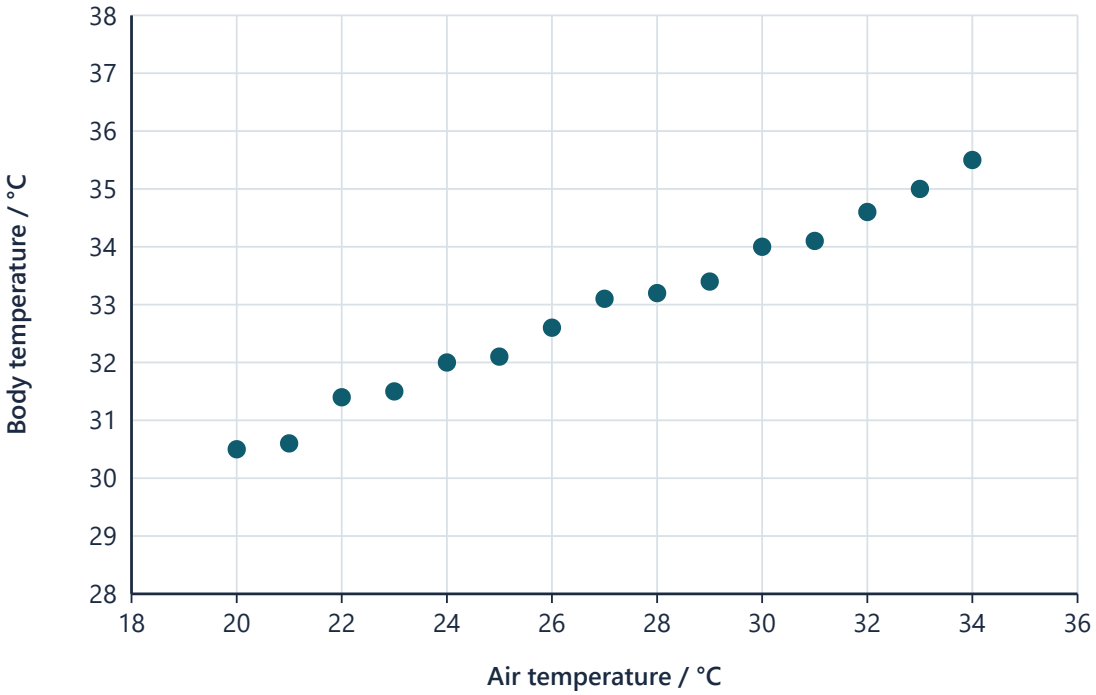
c EXPLAIN why high humidity increases the risk of overheating. [1]

d SUGGEST two ways runners can reduce the risk of heat illness. [2]

3 Body temperature in a sloth

6 MARKS

Brown-throated three-toed sloths (*Bradypus variegatus*) live in trees in Central and South America and eat leaves, which give little energy. The body temperature of sloths was measured at different air temperatures.



a **DESCRIBE** the relationship between air temperature and body temperature. [1]

b **CALCULATE** the mean change in body temperature for each 1 °C rise in air temperature between 20 and 34 °C. [1]

c **COMPARE** the sloths with a human in the same conditions. [1]

d **SUGGEST** two reasons why it is an advantage for sloths not to keep a constant body temperature. [2]

e **SUGGEST** why sloths eat more on warm days. [1]